

Artificial Intelligence, Inequality and Inclusion in the Middle East and North Africa

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In this lecture, Rizk unpacks the data ecosystem in the Middle East and North Africa, highlighting the unique demographic context of the region and its role in shaping the adoption and implementation of data and AI-based technologies. She begins by providing an important background around employment, poverty and inequality in the region before highlighting challenges around data infrastructure and technologies. Rizk later outlines rays of hope for data and AI use in the region. She concludes her lecture with a call for stakeholders to leverage AI and data to promote equality and inclusion.

Lecture Transcript

0:03 Good morning, afternoon, evening. My name is Nagla Rizk. I am Professor of Economics and the founding Director of the Access to Knowledge for Development Center at the American University in Cairo's School of Business. The title of my module today is: “Artificial Intelligence, Inequality and Inclusion in the Middle East and North Africa”, for short “AI, Inequality and Inclusion in MENA”. The purpose of my talk is to highlight the challenges that await the adoption of AI and data in the region, but also the opportunities, rays of hope and how we can capitalize on these for mitigating inequality and promoting inclusion. My talk is comprised of five sections. I will start by a brief background on the region, the context of the narrative that I'd like to present, then I will talk a bit about AI in MENA, data, infrastructure, people. I will highlight challenges, complexities and tensions, rays of hope. And then I will talk a bit about our work at the Access to Knowledge for Development Center.

1:10 The region is by no means homogeneous. It is a diverse region with variations in the level of resource richness and abundance of labor. So you'll find countries for example, like Saudi Arabia and Libya that are resource rich, but import the labor. And countries like Egypt, my own country, Morocco, and Jordan, for example, that are resource poor, but labor abundant. So when we, the impact of data, AI and technology are different depending on the context that we are studying. Nevertheless, the region has a number of commonalities. Most importantly, it is a young region. There is a youth bulge. You have the youth population, the highest in the world, 60% are less than 25 years of age, so there is the future for you. And when you look at projections for 2050, more and more young people are expected to be part of the population. In fact, some numbers between, aged, between 15 and 39, youth they comprise more than 40% of the total population. And in 2020, there were about 18 million more adults aged under 40 in the region than when the Arab Spring began.

2:27 These numbers are important. When we look at work, the MENA region has the highest rates globally of youth, share of youth unemployment, more than 27%. When we look at youth aged between 15 and 24, not in employment, not in education, not in training. In 2019, they are among the highest in the world. In 2016, about

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40% of university graduates in MENA were unemployed. Now look at women, the region comprises the lowest rate of female labor participation globally, and the highest rate of female unemployment globally. These numbers are important. Now, when we look at informal economies, again, this is the share of informal work in total employment globally, the region has high percentages of informal workers ranging between 50 and 70%.

3:28 As for poverty, you will find the poverty rates, this turquoise line shows poverty rates, yes, lower than some other parts of the world. However, the rates have been increasing. And when we look at the chart in front of us stopping just before 2019, of course, after COVID and the Ukraine war, now we look at numbers of people living in extreme poverty 5.5 dollars a day at purchasing power parity, the numbers are increasing. And in 2019, the number was 176 million, in 21, more than 190 million, and the World Bank projects that with every increase in food and energy prices could raise the number of the poor by more than 20 million in the region. Egypt alone, the most recent poverty rate was almost 30%. As for inequality, again, there are high rates of inequality in the region. If you look at the MENA region, the top 10% of the population are actually have a share almost 60% of the income. So we do have a region that is highly, you know, that has high rates of inequality, high rates of unemployment of the youth, women and the educated, and high rates of poverty.

4:49 Not only that, but inequality is multifaceted, complex layered by income, gender, ethnicity, social background, education, health services, digital access and use, employment, living conditions, political participation, geography. It is inequality of opportunity, not just of wealth. For example, this is a picture of, a random picture of, neighborhoods in Cairo. You have the contrast, the extremes from modern buildings to, you know, a low cost, you know, informal housing, actually slums. And you see similar pictures, actually, in countries like Saudi Arabia, which are supposed to be the richer countries of the region.

5:30 As for technology, let's take a look at the MENA region. This is the number of Internet users as percentage of the population. Yes, there has been an increase. Yes, it's higher than other parts of the world. But a closer look will also show us that this is more for richer countries of the region, the Emirates, Kuwait, Qatar, Saudi Arabia, Oman, as for in comparison to the rest of the countries. So there is a digital divide between countries of the region, not just within. Again, looking at mobile cellular subscriptions, you will find the MENA region doing relatively well, globally, indeed, higher than world averages. But these numbers are also mostly for the richer countries of the region, as you see on this diagram. So we also have a digital divide between and within countries by income, by age, by gender, education, and geography. And this is the context where technology comes in.

6:29 Now, as well, a word on freedoms in the MENA region. If you look at indices of economic freedom, you will find that the region relatively fares better than world averages, as far as economic freedoms are concerned. And this is mostly coming, again, from the richer countries of the region, the Gulf countries, the resource abundant. And this comes in comparison to what the region, looking at indices of democracy and political participation, you will find the region doing much less than world averages. This contrast between economic freedoms and civil liberties was highlighted as early as 2010. I had the good fortune to be a contributor to this report in 2009. And one of the conclusions of this report was that change in the actual situation of freedoms in the region in recent years has been confined to an improvement in economic freedoms. An analogous improvement in political and intellectual freedoms, democratic pursuits and freedom of expression has not occurred in most Arab countries. That was published in 2009. And of course, we know the result, the failure of trickle down, and the rest is history, no pun intended. The Arab Spring, taking place in 2011. At that time, about 250 million people out of

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the 400 million, or two thirds of the population, were classified as poor or vulnerable. So this is what happened. This is the context. And the context is important. Context matters when we look at technology. We all know that technology has the impact of, you know, exacerbating inequalities, marginalizing, but it can also be turned around for the pursuit of inclusion and for mitigating but also overcoming these inequalities and for promoting and advancing inclusion and development.

8:28 So let's take a look at AI in MENA. Let's take a look at data, infrastructure and people. Now, first data, of course, is the mine you know. There is, it is, you know, what we see in the region are a number of observations. First what I call the data lock by the state, because in many cases, data is not open. Data is behind barriers. Barriers could be political, could be technical. It is not available in a usable format. As well, you can have also a data asymmetry, inequality between the owners of data, of course, large multinationals, as opposed to young, homegrown businesses, within the different countries of the region. And of course, data is power. And this presents a barrier against small companies. Sometimes data is owned by large, even national corporations, not necessarily multinational. And in that sense, it presents a barrier against small companies. In one interview I had during my research on the subject, it was clearly stated that health data in particular, and that was before COVID by the way, that big labs sit on a wealth of health data that if only made available, of course, following the data protection, the proper legal landscape, this would create tremendous entrepreneurship opportunities for young businesses but also would improve health care for citizens. In fact, the court said that it would anticipate pandemics or actually the word was epidemics. It was before COVID. And this is quoted in my research.

10:08 So, we do have data challenges in the region. Then what I call is the data inaccuracy, a blur, you know. You will find data, there is a blur in data, because for many, for a number of reasons, the data may not reflect the reality on the ground. And invisibilities in data or data blurs will result in invisibility of communities from national policy making. For example, people living in informal housing that are not part of the urban planning schemes, that are not properly documented. We, I mentioned earlier, informal workers that are not part of the national data landscape will just be also invisible to national policy making and social protection. So these are images from housing in Cairo. One third of housing in Cairo, of course, we have modern buildings, but we do have one third of our housing are informal. We also have in Morocco, for example, a percentage of 23%. And also, in Lebanon. These communities exist in different parts of the region. And if they are not properly seen and documented in the data, they will be outside of the radar of national policy making as well.

11:19 One of the problems is what I call data myopia. In the sense that using indicators that are top down, that do not really see reality, that do not properly assess reality on the ground. And so the result is just a very myopic view of the reality. So this translates to policies that are not accurate and do not reflect what is happening on the ground, and policies that do not address the needs of the people. Again, using the Gini coefficient, as an example, is a failure. And again, I cite the Arab Spring, because the indices at the time were, you know, high rates of economic growth, the numbers were measuring income in the countries of you know, Egypt and Tunisia and the others. And clearly, this was a failure of data as well. Because the reality, the real inequality was not properly captured, using the metrics or the statistics that are the former ones that are usually designed in an office and do not take, are not creative enough to capture what is happening on the ground.

12:25 When we look at the infrastructure, we look at data storage and computing, that's another divide. Because there is an uneven access to data storage facilities between countries and within countries. It's also related to the

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digital divide. And even access to computing capacity, limited internet connectivity, all of this will affect the ability of adoption of the technologies. Again, even use of the cloud, there are the threats of being restricted to a specific vendor of data being, some people are skeptical of their data being stored with a specific vendor. And again, the weak power, weak infrastructure, is itself a challenge and can cause more regional digital inequalities with companies moving to either richer countries of the region or relocating completely in Europe or elsewhere in the Global North.

13:21 So these are issues that affect the adoption of AI in the region. The data issues are challenging, the infrastructure is challenging, but what about the people you know? Again, who designs the technology, who owns the technology? Who benefits? Who designs the algorithm, you know. Where are the beneficiaries of AI, you know? There are black boxes, the technologies are the black boxes within black boxes, and people do not have agency or ownership of what is going on there, you know. Who writes the algorithm? Are ready made algorithms? What data is being used? So again, this can challenge and amplify biases that can create, cause and create harmful consequences, especially to marginalized groups. So again, this is a problem between the biases, the inaccuracy of the data, the invisibility, the blur, biases and algorithms, all of this can cause allocative harms, especially if they're decontextualized and lacking the community and the domain experts.

14:28 What about work? People talk about jobs lost, jobs gained. Again, the skill structure is integral here of the region and we have seen statistics on unemployment of the university educated, so there is definitely a need for upskilling for these graduates for skills of the future. There is a gap, a clear gap there. And a very good example is call centers. In Egypt, for example, it was known for providing work opportunities for its youth who are typically graduates of universities and have skill with languages. Now, if this is going to be replaced by AI, these young men and women are going to be out of work and it has started already. So upskilling is very, very important. I also use the word cross skilling, because we need to skill between specializations and domains and technologies. So data scientists and technologists should also, if they are working in the health field, they need to understand about the dimensions of the health sector, but also people in the health sector need to learn the technologies. So this is what I mean by cross skilling. Again, in order to be able to gain new jobs and not be outside, you know, the labor market.

15:49 Of course, there is a challenge of retention and brain drain, and this is a real problem on the ground. I have interviewed startups in this field, and they have a challenge holding on to their human capital. What happens is that, you know, there is a brain drain either to the richer countries of the region, typically the United Arab Emirates, or to the Global North. Then, of course, there's a whole discussion about new forms of work, gig work, cloud work. The challenges, there of precarious work. The idea here of, you know, workers being, are not employed employees, and hence, like the benefits and the protection. But the irony is that, given the demographics that I spoke about earlier, you can see that these, this kind of work offers, indeed, opportunities, especially for cloud work that requires higher skills, and especially for these workers gaining foreign currency, because there is a clear policy, I mean, at least in Egypt and Lebanon, there is a very clear foreign currency shortage in the economy. So even with all the challenges that gig work, you know, on the ground and on the cloud, includes, it still can offer opportunities. So then the challenge becomes or the hope becomes, how can we have fair work practices in this form of work in a way to make it actually, you know, to capitalize on it for opportunities for the youth and for the women.

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17:16 Rays of hope. So I have outlined the challenges. Rays of hope is that the startup scene has picked up pretty well in the region. There's, you know, clear signs of active youth and women entrepreneurship, novel data collection methodologies, data driven innovation, and some government initiatives. Startup scene in MENA, here, for example, of course, United Arab Emirates tops the list. There's also of course, Saudi Arabia, but Egypt has been doing well, Jordan, Bahrain. Then also source the increasing rates of startups in the area. Egypt in particular has had a record high in 2021 in the VC deals and venture funding. Local startups have raised quite decent amounts of money, so this offers a great ray of hope for the region.

18:09 Another hopeful sign is novel data collection methodologies. Crowdsourcing, data driven innovation. HarassMap is an initiative, where, for the crowdsourcing for, you know, supporting women using technology to support women and protect them from harassment. Bey2ollak is another, also local startup, homegrown, for monitoring traffic in Egypt. Life is short to spend stuck in traffic. Cairo traffic is quite famous. So these are based on crowdsourcing data. And they offer solutions. Also, a positive sign, some governments have taken steps to, you know, adopt AI and to have strategies. So the countries on the map are countries that do have AI strategies, including Egypt, which I was fortunate enough to contribute to. And these are positive steps that are welcome.

19:04 And then this chart has, you know, sort of a summary of the environment around data. Still leaves, you know, quite a bit to be desired especially in the adoption of Open Government Partnership. We have only three countries in the region. Data protection, there are some steps you know, the circle that has those little teeth, this is sort of a draft law in Jordan. So we have some countries who do have data protection laws, right to privacy. Not all the countries but there are steps that are taking place. We still have quite a bit to go. Again with open, free and open source software in Egypt. Again, I was fortunate enough to draft the strategy for Egypt that was announced in 2014. So there are steps but then there is a lot to be desired, especially given the data locked by the state.

20:01 More tensions and complexities. The discourse over AI and inequality in the region is intertwined with its, with the region's unique political, economic and social context. That's why I spent time looking at this. The political economy of inclusion is very, very important in this part of the region, especially in light of its unique demographics. The dynamics of AI, and its impact on inclusion or inequality are embedded in these complexities. So, specifically, there are three tensions that shape the debate. First, the technology paradox. I mentioned it earlier. We all know that technology can lead to severe inequalities and exacerbating and marginalization, and so sort of leading away from the center, you know, centrifugal forces, but it can also bring centripetal towards the center, include, bring everyone in, you know, take steps for inclusion. And here is actually the challenge then becomes how can we, can AI serve these, you know, these trends? How can we use them to encourage inclusion, and definitely not only mitigate inequalities, or marginalization, but overcome them and transform the situation in a way that can promote inclusion?

21:18 I mentioned economic versus political freedoms. And this is, again, another tension. And this can, you know, we need to underline the gap there and to make sure that we use the technology for inclusion and for encouraging expression and civil liberties. And then there is also the threat of technological determinism. The idea that once we, you know, invest in technology, technology is, you know, a magic wand, it's the panacea. We should know that it's only, it may be necessary, but not sufficient. So, awareness of these tensions are important in understanding and addressing the dynamics, and making use of AI for inclusion in the MENA region. So, basically, you want to stay away from a top down approach that focuses on technologies that are not appropriate

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or that actually could further marginalize the underprivileged, and we want to be aware of these threats and to make every effort to actually invest in appropriate technologies that promote inclusion, and not marginalization.

22:31 If you will allow me, I'll say a few words about our work at the Access to Knowledge for Development Center. It is a multi, excuse me, multidisciplinary research. We are based at the School of Business, and we work with stakeholders from all over the world, locally, regionally, and internationally, on transformational research, evidence based research, advocacy to impact policy locally, and also to present the voice of the region in the global arena on these issues. These are my colleagues, this is my team at the center at the moment. We happen to be an all women team at the moment. And basically, we have a number of activities, I will just outline them. So we have the Fair Work Project in partnership with the Fair Work Network at the Oxford Internet Institute, and we look at Fair Work Practices in the ground gig economy, and how we can improve the conditions, work with platforms to improve the livelihood of workers and use this as an opportunity and mitigate the challenges so that workers are able to have decent work.

23:40 We have another project, a regional project, called New Work, Data and Inclusion in the Digital Economy. We lead regional research initiatives looking at not only gig workers as in the Fair Work project, but also we look at cloud workers. And this includes countries in other parts of, other countries of the region as I will show you shortly on the map. We have another project called Governing Responsible Data and AI in the MENA Region and here we are working with Birzeit in Palestine, Birzeit University. We lead the research track in different countries of the region looking at governance of data and AI, responsible data and AI in the MENA region. We are creating an observatory for that work that I will share with you in a minute. We are also the MENA hub for the A+ Alliance, the feminist AI research network looking at transformational AI to overcome inequalities, marginalization and actually change the design of AI and be you know, for inclusion of women and other marginalized groups. We do research on women and work informality in the post COVID-19 digital economy, and beyond in Egypt. And last but not least at the moment, we are engaged as part, we are the North African hub of the Open African Innovation Research Network, where we lead the research on alternative innovation metrics, the idea what I mentioned earlier about indices, looking at, you know, being assessed based on practices on the ground. So we try to look at innovation in Africa, not only through, you know, the sort of well known indices, mainstream indices, but we try to also create alternative indices that capture the innovation that takes place in Africa to provide accurate data on African innovation.

25:34 A bit more detail for examples. As I mentioned, governance of responsible data and AI in MENA. At the moment, we are leading a research track on governance of health data and AI in MENA region, in Morocco, Tunisia, Egypt, Jordan, Palestine and Lebanon. Then our next research track, will be food security. Both tracks are super urgent at this juncture for the MENA region. And as I mentioned earlier, we are creating the MENA AI Observatory, which will be launched pretty soon, and I will make sure to announce that for everyone so you can join the launch. So, this is with the support of IDRC and in partnership with Birzeit University. We're also, as I mentioned, the MENA hub for feminist AI research. For example, we are working with researchers in Jordan on a project looking at responsible algorithmic decision making to correct for biases in credit scoring for microfinance in Jordan, as I mentioned. And in Egypt, we are, we have also a fantastic research team, who's also proposed study that we are working with them on that on Explainable AI based tutoring system for Upper Egypt community schools and girls in this part of the world. It is their research and our role as the hub is to work with them, you know, on a research that becomes also hopefully a prototype and pilot.

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27:10 This is the research on new work, data and inclusion in the digital economy, a MENA perspective, with support from Ford Foundation. Again, we're working with partners in Morocco, Tunisia, Egypt, Jordan, and Lebanon, looking at specific case studies in every country, looking at Fair Work Practices and other ways of inclusion of gig work and/or cloud workers in the region in their respective countries. And we are doing our own study here in Egypt. This is the research on fair work in the gig economy. We have done two consecutive years two studies for Egypt. And we are very proud of the impact of the work, when sometimes the same platform, this research scores the fair work practices in the different platforms, and we're very happy to see platforms working with us from one year to the next. And their score improves because they're improving their practices with the workers as we collaborate together. And this is a win-win situation for everyone because this is also a fantastic opportunity for entrepreneurship for these platforms, for these startups who actually ended up being part of the dynamic startup scene and business entrepreneurship scene in Egypt.

28:24 And we also did a study, a regional study on practices, fair work practices in domestic service sector that is also based on the gig work model. Very quick examples. This is Egypt's AI strategy. And again, as I mentioned, very proud to have contributed to that. And this is, thanks to the wonderful work that my team, you know, it takes in collaboration with other parties in Egypt and elsewhere. So I'm very proud to be on the Technical Committee of the National Council, Egypt's AI policy, and this is really an opportunity to, you know, bring in what we, you know, what we're trying to achieve. Again, our work on Uber ride hailing study was quoted in the National Gazette when the Ride Hailing Act was issued in Egypt, in sort of, to improve the conditions of the Uber drivers. Then again, this is the national free and open source software strategy in Egypt announced in 2014. Examples of our global networks, you know, as I mentioned, Open AIR, Data for Development, Copyright^x, a Harvard Law School course that I teach at AUC. Of course, the Global Network of Internet and Society Resource Center, Access to Knowledge Global Academy, the Fairwork Network.

29:48 And examples of global partners. I will not read the names but just to give you an idea of the different partners that we have in the region, in Africa and globally. We're very proud partners of each and every one of them. One example of global activities that is very close to my heart is being part of the Executive Committee of the Global Symposium on AI and Inclusion in Rio de Janeiro with ITS, and the Network of Centers. That was November 2017. And, you know, I zoomed in, and there I am, and that is a picture with wonderful partners from all over the globe. And I think that was also a very good educational exercise for me.

30:30 Very quickly, our contributions to global publications, in fact, the chapter on AI and inequality in the MENA region at the very center of this slide is the base for this talk, you know, with of course, this was published in 2020. So I have updates for you today and you know, plenty of other stuff. And then participating in the Peace Paris Forum's report that came out earlier, in March of last year. Participating in the State of Open Data, writing the MENA chapter and then an update again, that's coming out very soon.

31:05 You know, a number of gender data, and evidence and gender equality. Again, writing the MENA chapter here, doing research on women drivers in ride sharing, doing the chapter on who owns the media in the MENA region as part of the book by Professor Noam. And this is just, you know, a number of publications that we have on this topic.

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31:29 I guess, in the end, the final word that I want to say is that at the Access to Knowledge for Development, what we really believe in is the role of knowledge technology for development, hence, it's our namesake. So with all the changes that are happening around us, it is very important to be very conscious of the ethics of AI and to make sure that we use the technologies, as the technology as transformational in making sure that we mitigate inequalities. Not only mitigate them, but also overcome them and advance inclusion in MENA for everyone, not only the marginalized, but for everyone in this region that has a very unique demographic and, you know, labor market characteristics. Thank you very much.